AI Advancements, Future Trends, and Regulatory Comparisons

Module 2 Assignment | ITAI 2372

(Because AI waits for no one — not even regulators)

Part 1: Three Recent AI Advancements

If you blinked between July and September 2024, you probably missed three major AI breakthroughs. Do not worry, this essay has you covered. With the enthusiasm of a puppy and the rigour of a slightly caffeinated student.

1. Open-Source Models: LLaMA 3 and Mistral 7B — Democratising the Brain

Meta's **LLaMA 3** and Mistral's **Mistral 7B** represent a pivotal shift: world-class AI capabilities no longer locked behind expensive API paywalls. LLaMA 3 is Meta's most capable open-source large language model to date. The open source offers competitive performance with closed proprietary models across reasoning, coding, and multilingual tasks. Mistral 7B is the star efficiency of the show. It punches well above its weight class, delivering strong results with far fewer param than its rivals. This makes it deployable on consumer hardware.

Why does this matter? Because open-source AI is to AI research what the printing press was to literacy. Researchers at universities, startups in developing countries, and independent developers can now build, fine-tune, and experiment with frontier-grade models without needing the budget of a small nation. Innovation accelerates when gates are removed.

> 🔑 **Key Insight**
> Open-source models like LLaMA 3 and Mistral 7B are redistributing AI power from a handful of tech giants to the global developer community — a trend with enormous implications for who shapes AI's future.

2. Multimodal AI: Gemini 1.5 Pro and Claude 3.5 Sonnet — AI That Can Actually See

Google DeepMind's **Gemini 1.5 Pro** and Anthropic's **Claude 3.5 Sonnet** pushed multimodal AI . The ability to process and reason across text, images, audio, and video. Allowing simultaneously to new heights. Gemini 1.5 Pro introduced a one-million-token context window which allows it to process entire codebases, lengthy documents, or hours of video in a single pass. Claude 3.5 Sonnet made significant strides in vision tasks and coding.The method to outperform many rivals on software engineering benchmarks.

Think of it like upgrading from a phone that can only call, to one that calls, emails, takes photos, and translates menus in real time. Multimodal AI is not just a technical upgrade. It opens entirely new application categories, from automated medical image analysis to AI-assisted creative production.

> 🔑 **Key Insight**

> Multimodal AI collapses the boundary between different types of information. A model that can read a document, analyse a chart, and discuss both in natural conversation is qualitatively different — and far more useful — than one that can only handle text.

3. OpenAI's o1 Series — Teaching AI to Think Before It Speaks

OpenAI's **o1 model series** introduced a novel approach: rather than immediately generating a response, o1 performs extended internal 'chain-of-thought' reasoning before answering. This produces dramatically better results on complex reasoning tasks such as mathematics, logic puzzles, and scientific problem-solving. That trip up conventional models.

The analogy is simple: compare a person who blurts the first answer that comes to mind versus one who pauses, thinks through the problem systematically, and then responds. o1 is the latter. It achieved near-PhD-level performance on the GPQA benchmark for scientific reasoning and ranked in the 89th percentile on competitive programming tasks. The results that were essentially science fiction for AI just two years prior.

> 🔑 **Key Insight**
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> The o1 series signals a shift from simply scaling model size to improving reasoning quality. This architectural innovation may prove more significant long-term than raw parameter counts.

Part 2: Australia vs the EU — A Regulatory Showdown

Comparing Australia's approach to AI regulation with the European Union's is a little like comparing a backyard barbecue to a formal seven-course Michelin-star dinner. Both aim to feed the guests. Well their methods, timelines, and dress codes are wildly different.

Australia: 'She'll Be Right, Mate' (Voluntary & Principles-Based)

Australia's AI governance is anchored by its **AI Ethics Framework**, published in 2019, which outlines eight voluntary principles: fairness, reliability, privacy, transparency, contestability, accountability, human-centred values, and inclusiveness. Note the key word: **voluntary**. There is no binding law compelling organisations to follow them.

The Australian approach reflects a philosophy of enabling innovation first, regulating second. The government has signalled intent to strengthen AI oversight. Particularly following the 2023 Safe and Responsible AI consultation. However, comprehensive legislation remains in development. In the meantime, existing laws (consumer protection, privacy, anti-discrimination) apply to AI where relevant, creating a patchwork rather than a purpose-built framework.

- **Strength:** Flexible, innovation-friendly, low compliance burden for businesses.
- **Weakness:** Voluntary frameworks are only as good as the goodwill of those following them — and the AI industry has not always had an abundance of goodwill.

The European Union: 'Show Me Your Papers' (Mandatory & Risk-Based)

The EU took the opposite approach. The **EU AI Act**, which entered into force in August 2024, is the world's first comprehensive, legally binding AI regulation. It classifies AI systems into four risk categories: Unacceptable Risk (banned outright — social scoring, manipulative AI), High Risk (strict requirements — medical devices, hiring tools, credit scoring), Limited Risk (transparency obligations), and Minimal Risk (largely unregulated).

High-risk AI systems must undergo conformity assessments, maintain technical documentation, ensure human oversight, and be registered in an EU database before deployment. Non-compliance carries fines of up to 7% of global annual revenue. A number that tends to get the attention of even the largest tech companies.

- **Strength:** Clear, enforceable standards that protect citizens and create legal certainty.
- **Weakness:** The law took years to pass and risks being overtaken by the very technology it seeks to govern before enforcement even begins.

The Comparison at a Glance

Aspect	Australia	European Union
Approach	Voluntary, principles-based	Mandatory, risk-based law
Key Document	AI Ethics Framework (2019)	EU AI Act (2024)
High-Risk AI	Encouraged best practice	Strict pre-market requirements
Enforcement	Self-regulation + guidance	National regulators + fines up to 7% of global revenue
Speed	Flexible & fast-moving	Thorough but slow (years to pass)
Vibe	Laid-back: 'Give it a fair go'	Rigorous: 'Prove it or don't ship it'

The fundamental tension is this: Australia prioritises **speed and flexibility** while the EU prioritises **safety and accountability**. Neither is purely right. Australia risks under-regulating harms that are already occurring. The EU risks over-regulating in ways that push innovation outside its borders. The ideal approach likely borrows from both. The EU's risk-tiered rigour applied with Australia's adaptability and speed.

Part 3: Two Future Trends to Watch

Trend 1: Agentic AI — From Chatbots to Co-Workers

The next major AI evolution is not simply making models smarter. It is making them **autonomous agents** capable of taking multi-step actions in the real world. Rather than answering a question, an AI agent books the meeting, writes the follow-up email, and updates the project tracker. Without being asked to do each step separately.

Tools like OpenAI's Operator and Anthropic's computer-use capabilities are early signals of this shift. Agents will browse the web, write and execute code, interact with software interfaces, and manage long-horizon tasks. The economic implications are profound and so are the regulatory ones.

Both Australia and the EU will need to address a fundamental question: when an AI agent takes an action that causes harm, **who is liable?** Current frameworks assume a human made the decision. Agentic AI breaks that assumption entirely. The EU AI Act's existing accountability requirements will need significant extension to cover autonomous agent behaviour. Australia will need to move from voluntary principles to enforceable liability rules.

> 📋 **Regulatory Implication**
> Agentic AI demands clear liability frameworks. Voluntary principles are insufficient when an AI autonomously books a flight, fires an employee, or approves a loan. Both Australia and the EU face urgent work in this space.

Trend 2: AI and Scientific Discovery — The Accelerating Lab

AI is transitioning from a tool **used in** science to a participant **driving** scientific discovery. Building on DeepMind's AlphaFold protein-folding breakthrough. AI systems are now being applied to materials discovery, drug synthesis, climate modelling, and physics. Microsoft's AI-driven discovery of a new battery material and Insilico Medicine's AI-designed drug. Candidates entering clinical trials are early indicators of a profound shift.

The pace of discovery is set to compress dramatically. Problems that took decades may take years; problems that took years may take months. This creates both extraordinary opportunities (faster cures, cleaner energy) and regulatory challenges (how do you approve an AI-discovered drug when the discovery pathway itself is opaque?).

For regulators in Australia and the EU, AI-driven scientific discovery will require new frameworks around **explainability and validation**. If an AI system proposes a novel drug compound, regulators need assurance that the AI's reasoning is sound. Not just that the output passed a clinical trial. The EU AI Act already lists medical devices as high-risk. Extending this thinking to AI-accelerated drug discovery pipelines will be a priority area.

> 📋 **Regulatory Implication**
> AI-driven science will outpace traditional approval timelines. Regulators must develop 'AI-native' validation frameworks that can assess process transparency, not just clinical outcomes.

Conclusion

The period from July to September 2024 alone delivered three transformative AI advancements. Open-source democratisation, multimodal reasoning, and deliberate chain-of-thought AI. Each with significant implications for how we work, create, and govern.

The regulatory responses from Australia and the EU represent two philosophically distinct bets on how to handle a technology that is evolving faster than any regulatory process in history. Australia backs flexibility, the EU backs structure. The emerging trends of agentic AI and AI-driven scientific discovery will test both approaches, and may ultimately require a convergence. The EU's enforceable accountability paired with Australia's adaptive speed.

One thing is certain: the AI genie is not going back in the bottle. The only question is whether our regulatory frameworks are **wise enough to keep up**. On current evidence, we have our work cut out for us. In results, what can AI now do? Perhaps it can help us write the regulations too.

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